

Look Where It Changed: Robotic Next-Best-View Planning for Updating Gaussian Splatting

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1 Point Simulation

To conduct a preliminary study and test the camera pose optimization and planning algorithm, the problem is recreated in a simplified manner within a synthetic, controlled, and reproducible environment in which an arbitrary number of scene configurations can be generated procedurally. At this stage, the change detection component of the pipeline is assumed to be given and to work perfectly; therefore it is replaced by a proxy. Only the components operating on the point cloud are reproduced, while the photometric channel is not simulated.

1.1 Synthetic Point Cloud

A real depth sensor (RGB-D) or a structure from motion (SfM) outputs only the reconstruction of the scene’s surfaces; it cannot provide the points contained inside the volume of the present objects. This has direct consequences for reconstruction properties, such as occlusions, incident angles, and the boundaries of known and unknown zones. To simulate this, it is necessary to first generate the surface of a random solid volume and then sample its points.

The solid is defined implicitly by a scalar field $g(x, y, z)$, which associates a number with each point in 3D space:

- $g(x, y, z) < 0$ point lies inside the solid volume;
- $g(x, y, z) = 0$ point lies on the surface of the volume;
- $g(x, y, z) > 0$ point lies outside the solid volume.

This representation simplifies the evaluation of point positions respect to the volume and the computation of boolean operations. The union of two solids is the minimum

of their fields and their intersection is the maximum [1]. A procedural level surface $g(x, y, z) = 0$ can be defined as:

$$g(x, y, z) = u - f(x, y, z) - m(x, y, z), \quad (1)$$

where:

- u is the level that fixes the size of the solid;
- $f(x, y, z)$ is a random field;
- $m(x, y, z)$ is a mean function decreasing towards the borders of the scene.

1.1.1 Random Field

The synthetic scene is used to challenge the planning algorithm; this requires the random field $f(x, y, z)$ used to build the surface to satisfy five properties:

1. Genuine variety of achievable shapes. Concavities, cavities and all the separate components have to be generated randomly to test the planner in the most neutral conditions possible.
2. Tunable difficulty of the test using interpretable parameters. It should be possible to control the size of the folds to change the level of self-occlusion in the scene and on the surface roughness to influence the reliability of the visibility filter.
3. Absence of privileged positions and directions. The result should not be determined by the reference frame, unless explicitly imposed.
4. True function, valuable together with its gradient in any point. The visibility ground truth, the sampled points projection onto the surface, and the measurement of the changed area all query the field at arbitrary points, and all of them have to see exactly the same surface.
5. Known distribution of values. The expected size of any object in the scene should be fixed by solving an equation rather than manual tuning.

It is possible to satisfy all these requests defining a gaussian process, where in every i -th point $f(x_i, y_i, z_i)$ is a gaussian and its similarity to $f(x_j, y_j, z_j)$ with $i \neq j$ depends exclusively on the distance between (x_i, y_i, z_i) and (x_j, y_j, z_j) via a kernel function.

References

- [1] A. Ricci, “A Constructive Geometry for Computer Graphics,” *The Computer Journal*, vol. 16, no. 2, pp. 157–160, Feb. 1, 1973, issn: 0010-4620, 1460-2067. DOI: 10.1093/comjnl/16.2.157 Accessed: Sep. 23, 2026. [Online]. Available: <https://academic.oup.com/comjnl/article-lookup/doi/10.1093/comjnl/16.2.157>